Questions

Population Genetics (PP)

HWE & F statistic

From a case-control study, the following genotype counts have been observed for a SNP with alleles A and B:

Cases:	$n_{obs}(AA) = 159$	$n_{obs}(AB) = 122$	$n_{obs}(BB) = 19$
Controls:	$n_{obs}(AA) = 120$	$n_{obs}(AB) = 139$	$n_{ob}\ (BB) = 41$

Test this SNP for deviation from Hardy-Weinberg equilibrium, calculate the F_{ST} statistic in controls, test for association under a genotypic and under an allelic risk model and calculate the odds ratios for the SNP genotypes and alleles, respectively.

I. Testing for deviation from Hardy-Weinberg equilibrium (HWE) in controls

1. Calculate the genotype and allele frequencies in controls:

$n = n_{obs}(AA) + n_{obs}(AB) + n_{obs}(BB) =$ ....

$f_{obs}(AA) = n_{obs}(AA) / n =$ ....

$f_{obs}(AB) = n_{obs}(AB) / n =$ ....

$f_{obs}(BB) = n_{obs}(BB) / n =$ ....

$f(A) = [2\times n_{obs}(AA)+n_{obs}(AB)] / 2n =$ ....

$f(B) = [2\times n_{obs}(BB)+n_{obs}(AB)] / 2n =$ ....

2. Calculate the expected genotypic counts under the null hypothesis of HWE:

$n_{exp}(AA) = n \times [\ f(A)\times f(A)\ ] =$ ....

$n_{exp}(AB) = n \times [\ 2\times f(A)\times f(B)\ ] =$ ....

$n_{exp}(BB) = n \times [\ f(B)\times f(B)\ ] =$ ....

3. Arrange observed and expected genotype counts in a 2×3 table and calculate the chi-square statistic:

	AA	**AB**	**BB**
Observed (n_{obs})	120	139	41
Expected (n_{exp})	....	....	....

$$X^2 = [n_{obs}(AA)- n_{exp}(AA)]^2 / n_{exp}(AA) + [n_{obs}(AB)- n_{exp}(AB)]^2 / n_{exp}(AB) + [n_{obs}(BB)- n_{exp}(BB)]^2 / n_{exp}(BB)$$

= .... + ... + ...

= ....

4. Obtain the corresponding *P*-value from a *one*-df χ^2 distribution:

Quantiles of the 1-df χ^2 distribution using R:

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pchisq( <<QUANTILE>>, df=1, ncp=0, lower.tail=F)
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p = ...

II. Calculating the F_{ST} statistic in controls

The F_{ST} statistic can be calculated by the formula given below (introduced in the lecture on population genetics). Use the control frequencies calculated in Exercise 1 on HWE testing.

$$F_{ST} = [\, f_{obs}(AA) - f(A)\times f(A)] \quad / \quad [\, f(A) - f(A)\times f(A)\,]$$

F_{ST} = ... / ....

F_{ST} = ...

Questions

1. Is there statistical evidence at the 0.05 level that the marker is not in HWE?

2. The reported genotype counts were observed in controls only. Would it be beneficial to merge the control genotype counts with those from the cases to test HWE testing, since it would increase the sample size and power for this test? Give a reason for your answer.

3. How do you interpret this value of F_{ST} with regard to the sample (see the lecture for an interpretation of the value)?

Question 7

You are performing a study using the UK Biobank and for your phenotype of interest you have 50,000 cases and 100,000 controls. For a disease with 10% prevalence, disease allele frequency of 0.01, where each variant has an effect size of 1.2 under a dominant model what would be the power for an aggregate test where the cumulative allele frequency is 0.01 _________and a single variant test ____________? Clue use the appropriate alpha for each test.

Question 8

Using have a replication sample of 50,000 cases and 50,000 controls and you plan to try to replicate 15 genes and 100 variants. Using the same parameters as in question 7 what would be your power to replicate a.)________________? Note for alpha use a Bonferroni correction.

Question 9

For the above power calculations, you have been using the relative risk which only approximates the odds ratio when a.) _______________________? You are performing a power calculation for a case control study for a disease/variant frequency of 0.01. You use a dominant model and a gamma of 1.2 for a disease with a prevalence for 0.2. What is the odds ratio for which the power calculations are being performed b.) _________________? *Use Genetic Power Calculator – information not provided by GAS or Power2.

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